

CMPS 04

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1 Dynamics

Why are **differential equations** an important technique for computational modelling in psychology and neuroscience?

Spiking neuron models rely on differential equation to capture the evolution of voltage as a function of time. Dynamics is the study of things changing over time, and neural dynamics gives us important insights into brain and neuronal activity normally and as a consequence of disease.

An interest in dynamical systems has been a part of computational neuroscience since the days of Hodgkin and Huxley (more on them later). But with the development of cheaper and more powerful computing capacities their use as expanded practically and theoretically. It is increasingly feasible to simulate complex models in high dimensions. This advance parallels the use of multiple electrode recordings that yield high dimensional data.

To ground our understanding with the use of differential equations in computational neuroscience we will produce simple, single neuron simulations of the integrate and fire variety. An excellent and concise summary of these ideas, one that I drew upon heavily for this course is.¹

1.1 Beginning to Think Dynamically

Assume you have a derivative that is equal to $x - x^3$ (see Figure 1). What would it mean to "solve" this equation?

What you want is some *function* of the variable x that will equal what you get when you take its derivative. For this function that is easy enough.

¹David Terman, "An Introduction to Dynamical Systems and Neuronal Dynamics," in *Tutorials in Mathematical Biosciences I* (Springer Berlin Heidelberg, 2005), 21–68, https://doi.org/10.1007/978-3-540-31544-5_2.

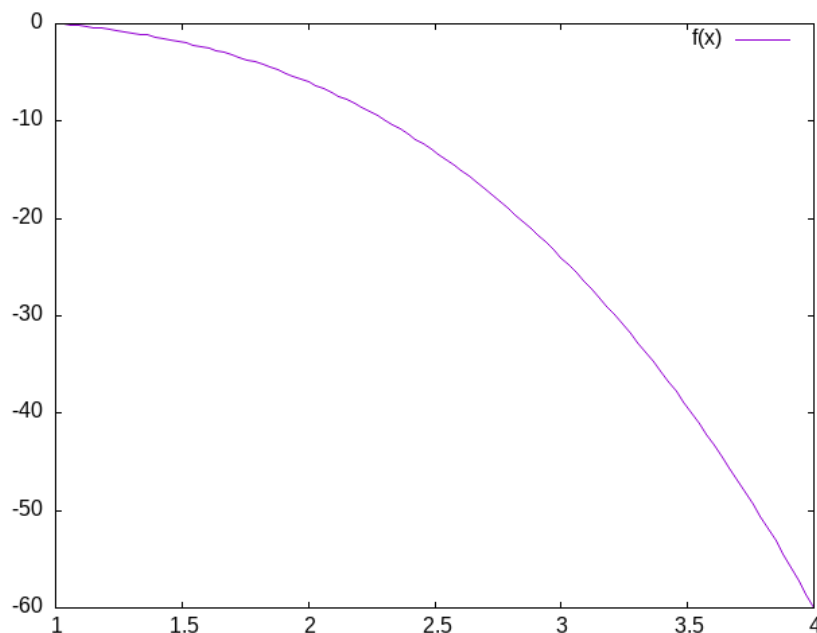


Figure 1: A plot of $f(x) = x - x^3$

You simply integrate. But what if we think of x as dynamic. It is not fixed. It varies with time, like the voltage inside a neuron, the air temperature, or the strength of a memory trace. Then your equation becomes, $\frac{dx}{dt} = x - x^3$. Integrate that.² In this case we can find an analytical³ solution. But we don't have to do much to our equations to make such solutions very difficult or impossible. For most of the use practical use cases in computational neuroscience we do not solve these equations analytically. Rather, we use the computer to approximate their behavior. Most of the equations in neuroscience, and biology generally for that matter, are well behaved enough that we can get away with this.

We will now review the action potential and lay the ground work for writing our own code to simulate a neuron's firing.

1.2 The Action Potential

²It is not *that* hard, and Claude.AI (as of <2024-12-19 Thu>) can do it for you. But it is substantially harder than before.

³What does "analytical" mean here?

How is an action potential like a flush toilet?

- 10 minutes to brush up on what an action potential is.
- Then be able to draw one.
- What are the axes?
- What ion causes the upward deflection?
- What causes the repolarization?
- Who discovered the action potential?
- Who won the Nobel Prize for characterizing the ionic events of the action potential experimentally and building a mathematical model?

1.3 A Mathematical Aside

I often find that psychology students let mathematical notation get in their way of understanding. The unfamiliar symbols lead people to skip the most important part of computational papers. The most important thing to learn is that mathematical notation is just like an emoji: a concise way to express a more complex idea. Once you learn the "slang" you can look beyond the characters. Here is an example of how simple math can start to look forbidding when notated.

What does $\sum_{x \in \{1,2,3\}} x = 6$ really mean?

Another confusing feature of mathematical notation is that there are often many ways to say the same thing and different mathematical communities have different conventions. But just because you cannot speak French you do not assume the French are really saying something different. So, when faced with a strange mathematical expression be open to the idea that it is just another way to say something you are already familiar with.

For example, $\frac{dx}{dt}$, $\dot{x}$, x' , $f'(x)$ are all different ways of talking about derivatives and derivatives are at the heart of differential equations.

1.4 Derivatives are Slopes

1. What is a slope?
2. When in doubt return to definition.
3. Deriving the definition of a derivative.
4. What is the definition of a derivative?

1.4.1 Your computer is a tool for exploration

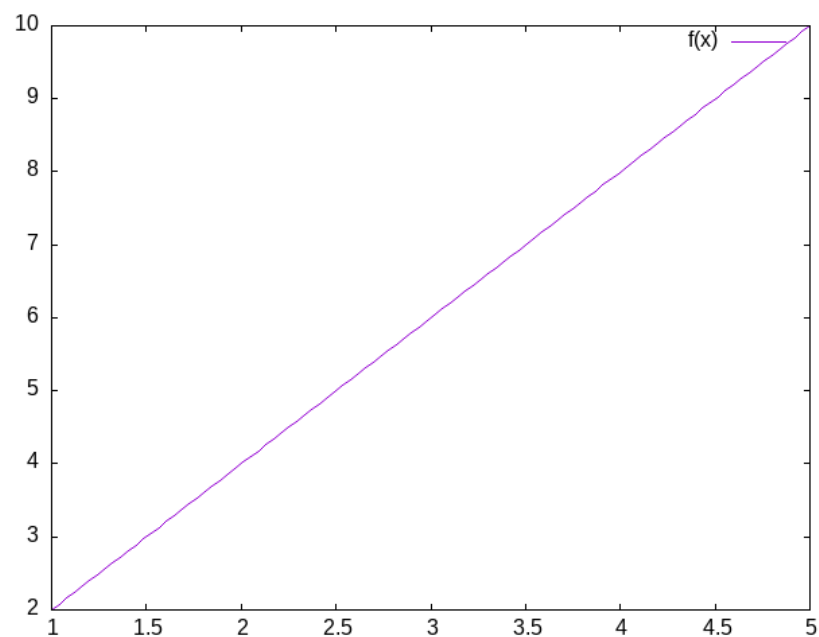


Figure 2: What is the slope of this line?

1.4.2 To approximate the derivative of a curve at a point draw a straight line close by

You pick two points that are "close enough" and you get an answer that is "close enough." If your answer isn't "close enough" then you move your points closer, until *in the limit* there is an infinitesimal distance between them.

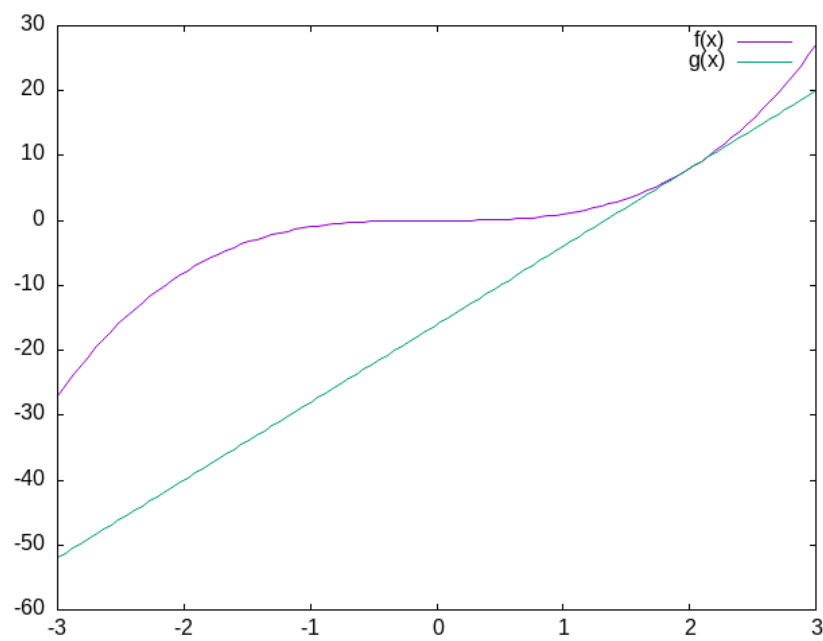


Figure 3: What is the slope of this curve? And what is missing from that question that you need to know to give an answer?

Definition of a Derivative

$$\frac{dx}{dy} = \frac{f(x+\Delta x) - f(x)}{x+\Delta x - x}$$

1.5 Digression:

To write math for publication you will probably need to know a little bit of $\text{\LaTeX}$. $\text{\LaTeX}$ versions can be found for all sorts of environments and operating systems. To get started try writing $\text{\LaTeX}$ in VS Code. A guide for word processor users can be found here. However, in the short term the easiest way to get started may be to use the free Overleaf account you have as a UWaterloo student. This is great for collaboration as it lets you share and edit a document jointly, like a google doc.

1.5.1 Using Derivatives to Solve Problems With a Computer

What is a square root? You might want to review our earlier exercise to see how the derivative is playing a role in our solution.

1.6 Practice Simulating With DEs**1.6.1 Frictionless Springs**

We want to code neurons, but to get there we should feel comfortable with the underlying tool or we won't be able to adapt it or re-use it for some new purpose. We will start with a very simple and concrete example: the frictionless spring.

$$\frac{d^2s}{dt^2} = -P s$$

where 's' refers to space, 't' refers to time, and 'P' is a constant, often called the spring constant, that indicates how stiff or springy the spring is.

Imagine that you knew this derivative. It would tell you how much space the spring head would move for a given, very small, increment of time. We could then just add this to our current position to get the new position and repeat. This method of using a derivative to iterate forward is sometimes called the Euler method.

Returning to our definition of the derivative:

$$\frac{s(t+\Delta t) - s(t)}{\Delta t} = \text{velocity} \approx \frac{ds}{dt}$$

But our spring equation is not given in terms of the velocity it is given in terms of the acceleration which is the second derivative. Therefore, to find our new position we need the velocity, but we only have the acceleration. However, if we knew the acceleration and the velocity we could use that to

calculate the new velocity. Unfortunately we don't know the velocity, unless ... , maybe we could just assume something. Let's say it is zero because we have started our process where we have stretched the spring, and are holding it, just before letting it go. We have asserted an *initial value*.

How will our velocity change with time?

$$\frac{v(t+\Delta t)-v(t)}{\Delta t} = \text{acceleration} \approx \frac{dv}{dt} = \frac{d^2s}{dt^2}$$

And we have a formula for this. We can now bootstrap our simulation.

Note the similarity of the two functions. You could write a helper function that was generic to this pattern of *old value + rate of change times the time step*, and just used the pertinent values.

How do we know the formula for acceleration? We were given it!

```
import localcode.spring as s
s.plot_spring()
```

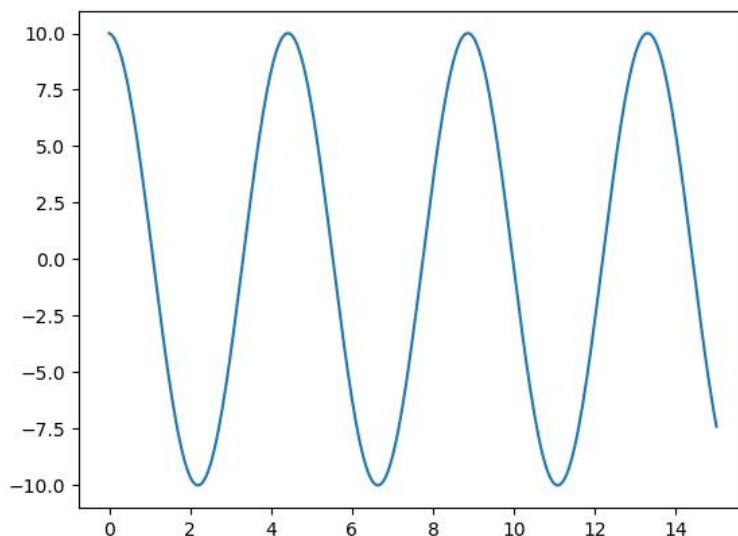


Figure 4: Spring constant of 2 with a starting position of 10.

1. Now You Try **Damped Oscillators**

Provide the code and a plot for the damped oscillator. It has the formula of

$$\frac{d^2 s}{dt^2} = -P s(t) - k v(t)$$

1.7 Dynamical Systems

Sometimes a variable can be a function. This is not literally true, but sometimes we think of a value like "x" as a location in a graph, but we know that in some situations the value of "x" may change with time. Thus we can also think of "x" as "x(t)" where the value of x in this case depends on time. This dynamical systems approach is the idea as we use to implement our spiking neuron models. We think of x as a function. Sometimes it may be time, but often it will be voltage.

1.7.1 Fixed Points

Give me a place to stand, and I shall move the world. —Archimedes (paraphrase)

Fixed points are the points in a function where it no longer changes. It becomes *fixed*.

What is the derivative at a fixed point?

We will use the ideas of fixed points, at least informally, to determine certain starting parameters for our Hodgkin & Huxley model. We will assume that the system evolved to some point in time where the derivative of a function with time was zero. Then we will infer our parameter based on that assumption.

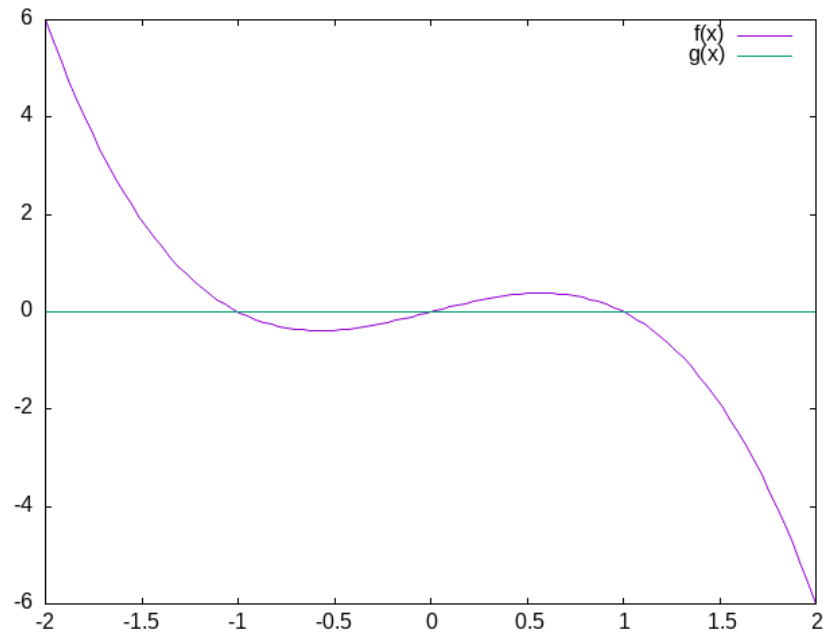
You can learn a lot about a system by looking at its fixed points

Class Exercise What are the fixed points for a function where $\frac{dx}{dt} = x - x^3$?

For the function above what are the fixed points and are they *stable* or *unstable*? What do you think is meant by the term *stability*?

1.7.2 Class Activity

Informally, you can assess stability by looking nearby the fixed points to see if they are pushed toward or away from the fixed point. Which of these three are stable or unstable?



2 Modeling the Action Potential

2.1 Integrate and Fire Neuron

In this section we take a look at the history and math of the computational model of neuron firing called "Integrate and Fire" (I&F). The I&F model uses math essentially the same as the spring example.

Is the integrate and fire model used much in modeling in the present time? answer

2.1.1 History of the Integrate and Fire Model

1. Louis Lapicque - Earlier Computational Neuroscientist
 - Modern Commentary on Lapicque's Neuron Model
 - Original Lapicque Paper (translated)
 - Brief Biographical Details of Lapicque
2. Lord Adrian and the All-or-None Action Potential One of the first demonstrations of the all-or-none nature of the neuronal action potential was made by Lord Adrian. Lord Adrian was an interesting



Figure 5: La Pique Laboratory at the Sorbonne Paris

scientific figure. He asserted that some of his prowess at electrophysiology stemmed from his interest in and training in fencing.

- (a) When was the action potential demonstrated?
- (b) What was the experimental animal used by Adrian?

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1420429/pdf/jphysiol01990-0084.pdf>

Want more details? There is an excellent free book available Spiking Neurons They also have another more modern book out too Neuronal Dynamics

While Hodgkin and Huxley provided the first robust computational model of the neuronal action potential their model is quite complex, as we will soon see. Is all that complexity necessary? That of course depends on the nature of the scientific question, and if we are primarily interested in whether a spike has or has not occurred, and not the ionic events that produced the spike, we may find our experimental questions dealt with much more concisely by a much simpler model of neuronal spiking: the leaky integrate and fire model.

The formula for the leaky integrate and fire neuron is:

$$\tau \frac{dV(t)}{dt} = -V(t) + R I(t)$$

In the next sections we will describe how this simplification came to be, and use it as the basis for learning some of the elementary electrical laws and relations upon which it is based.

2.1.2 Electronics Background

The following questions are the ones we need answers to to derive our integrate and fire model.

1. What is Ohm's Law?
2. What is Kirchoff's Point Rule?
3. What is Capacitance?
4. What is the relation between current and charge?

Ohm's Law (empirically observed): $V = IR$

What is the relationship between current and charge? Current: The derivative of charge with respect to time,

$$I = \frac{dQ}{dt}$$

What is **Kirchoff's Point Rule**

Current sums to zero: All the current sources going to a node in a circuit must sum to zero.

What is capacitance?

Capacitance is a source of current. A capacitor is a sandwich of two conducting surfaces with a non-conducting body in between. If you a charge to one side, the electrons gather there. They can't leap the gap, so they exert an attraction for particles of the opposite charge on the other side of the gap. If you suddenly stop the charge then charge races around and you discharge a current.

Explain the relationship, mathematically, between capacitance, charge, and voltage.

$C = Q/V$ The volume of charge, per unit area, divided by the voltage that produces this imbalance in charge.

What happens when you differentiate this equation with respect to time and treat the capacitance as a constant? $C \frac{dV}{dt} = \frac{dQ}{dt} = I$

1. Formula Discussion Questions To understand our formula clearly we should review the meaning of the key symbols and notation.

What does $\frac{dV}{dt}$ mean? What does $\frac{1}{\tau}$ mean? Why does the voltage term on the right have a negative sign? What is $I(t)$?

Answers:

It is the derivative. It is the how the voltage changes as a function of how time changes.

It is the membrane time constant and can be related to the membrane capacitance. Since it is a constant, with a clever choice of units you can assume it to be one and make it disappear.

To get the intuition of a model you don't always have to compute things. You can also get some qualitative behaviour just by looking at it. The larger the voltage the more negative becomes its rate of change and vice versa. It drives everything back to some point at which the rate of change to an equilibrium point. We will come back to this notion of a fixed point or attractor.

It is the current term. I is the common abbreviation for current. Why? I don't know, can someone help?

To derive our equation we need to put all these fact together.

We recall that just like we used the derivative to help us figure out where the spring would be some small increment of time into the future, we use the same approach to compute our future voltage. That future voltage will also include a term that reflects an additional current that we have "experimentally" injected.

Can you tell why, looking at the integrate and fire equation, if we don't reach the firing threshold, we see an exponential decay?

Deriving the IandF Equation

$$I = I_R + I_C \quad (a)$$

$$= I_R + C \frac{dV}{dt} \quad (b)$$

$$= \frac{V}{R} + C \frac{dV}{dt} \quad (c)$$

$$RI = V + RC \frac{dV}{dt} \quad (d)$$

$$\frac{1}{\tau}(RI - V) = \frac{dV}{dt} \quad (e)$$

a: Kirchoff's point rule,

b: the relationship between current, charge, and their derivatives

c: Ohm's law

d: multiply through by R

e: rearrange and define τ

2.1.3 Coding up the Integrate and Fire Neuron

Most of the integrate and fire implementation is conceptually and practically identical to the spring example. You assume a starting voltage (initial state) and then you update that voltage using the differential equation for how voltage changes with time $\frac{dV}{dt}$

There is one critical difference though. Unlike real neurons the Integrate and Fire neuron model does not have a natural threshold and spiking behavior. You pick a threshold and everytime your voltage reaches that threshold you designate a spike and reset the voltage.

What I added below is a strictly cosmetic amendment that changes the first value after the threshold to a number much higher than the threshold so that when plotted it creates the visual appearance of a spike.

Additional Activities:

Adding a refractory period to the Integrate and Fire model.

Try altering the input current to see how that affects the number of spikes and the regularity of their spiking.

Next, change the form of the input current to be something other than a constant. I suggest trying a sine wave.

2.2 Integrate and Fire Homework

The Integrate and Fire homework has two components. One practical and one theoretical.

Practically, submit an integrate and fire program. Do something to show that you had something to do with the coding and you understand what is happening.

Theoretically, look at this article (pdf) and tell me how you feel our integrate and fire model compares to these actual real world spiking data when both are give constant input. What are the implications for using the integrate and fire model as a model of neuronal function?